The first factor is the anchor; the second and third are SDDR Bayes factors that chain through the extended model, $\mathcal{H}_1(\gamma_x) \to \mathcal{H}_\gamma \to \mathcal{H}_1(\gamma_0)$, and cancel it from the final comparison. The algebraic structure is analogous to Eq. 17 of Fowlie (2024), who used the same posterior-to-prior density ratio to compute Bayes factor contours over phenomenological parameters in particle physics; the key difference is that Fowlie's framework varies parameters that enter the *likelihood* (e.g., particle masses) and compares nested parameter values directly, whereas here γ governs the *prior distribution* under $\mathcal{H}_1$ and the null hypothesis $\mathcal{H}_0$ generally lies outside the extended model, necessitating the separate anchor $\mathrm{BF}_{10}(\gamma_0)$.

This identity holds whenever the extended model $\mathcal{H}_\gamma$ adds a proper hyper-prior on γ without otherwise changing the model structure for any fixed γ; that is, $\mathcal{H}_\gamma$ reduces to $\mathcal{H}_1(\gamma^*)$ when γ is fixed at γ^*. The anchor point γ_0 must lie within the support of the posterior distribution, a condition guaranteed whenever γ_0 falls in the interior of $[\gamma_L, \gamma_U]$.

In the simplest case, $\pi(\gamma) = \mathrm{Uniform}(\gamma_L, \gamma_U)$, the prior ratio $\pi(\gamma_0)/\pi(\gamma_x) = 1$, and the formula reduces to

$$\mathrm{BF}_{10}(\gamma_x) = \mathrm{BF}_{10}(\gamma_0) \times \frac{p(\gamma_x \mid y, \mathcal{H}_\gamma)}{p(\gamma_0 \mid y, \mathcal{H}_\gamma)}. \tag{9}$$

Once the extended model has been fit, the entire sensitivity curve is available: the Bayes factor at any γ_x is the anchor value multiplied by a ratio of two posterior density ordinates. The key computational saving is that the extended model $\mathcal{H}_\gamma$ adds only a single parameter to the original model $\mathcal{H}_1(\gamma_0)$ and therefore has essentially the same sampling cost as a single standard fit, yet it replaces the K separate fits that would be needed under brute-force gridding.

In practice, the procedure has four steps. First, $\mathcal{H}_0$ and $\mathcal{H}_1(\gamma_0)$ are fit to obtain the anchor $\mathrm{BF}_{10}(\gamma_0)$ via bridge sampling (Meng & Wong, 1996), SDDR (Dickey, 1971), or any other method for computing marginal likelihoods (for a comprehensive review, see Diciccio et al., 1997; Llorente et al., 2023). Second, the extended model $\mathcal{H}_\gamma$ is fit; this model is identical to $\mathcal{H}_1(\gamma_0)$ except that γ receives a hyper-prior $\pi(\gamma) = \mathrm{Uniform}(\gamma_L, \gamma_U)$ covering the sensitivity range. Third, the posterior density $\hat{p}(\gamma \,|\, y, \mathcal{H}_\gamma)$ is estimated from the MCMC samples of γ (see Section 2.3). Fourth, $\mathrm{BF}_{10}(\gamma_x)$ is computed via (9) for all γ_x of interest. The total cost is one standard Bayes factor computation plus one extended model fit, dramatically cheaper than the K separate fits required by brute-force evaluation, especially since the extended model adds only a single parameter (γ) and typically samples equally efficiently as $\mathcal{H}_1(\gamma)$.

2.1 Sensitivity to Multiple Hyper-Parameters

The method extends naturally to multiple sensitivity parameters. For an informed Bayesian t-test of Gronau et al. (2020) with a normal prior distribution $\mathrm{N}(\mu_\delta, \sigma_\delta^2)$ on effect size, one may wish to vary both the location μ_δ and the scale σ_δ simultaneously. The extended model then places a joint hyper-prior (e.g., a bivariate uniform distribution) on $\boldsymbol{\gamma} = (\mu_\delta, \sigma_\delta)$, and the posterior density ratio is estimated from the bivariate posterior samples:

$$\mathrm{BF}_{10}(\boldsymbol{\gamma}_x) = \mathrm{BF}_{10}(\boldsymbol{\gamma}_0) \times \frac{p(\boldsymbol{\gamma}_x \mid y, \mathcal{H}_\gamma)}{p(\boldsymbol{\gamma}_0 \mid y, \mathcal{H}_\gamma)} \times \frac{\pi(\boldsymbol{\gamma}_0)}{\pi(\boldsymbol{\gamma}_x)}. \tag{10}$$

This produces a two-dimensional Bayes factor sensitivity surface analogous to those of Franck & Gramacy (2020). Because multivariate density estimation becomes harder with increasing dimension, the method works best for only a few parameters; for higher-dimensional sensitivity spaces, the Gaussian process surrogate approach of Franck & Gramacy (2020) may be preferable.

2.2 Bayesian Model Averaging

Importantly, the method also applies to Bayesian model averaging (BMA; Hoeting et al., 1999; Fragoso et al., 2018; Hinne et al., 2020), where inference is averaged across multiple component models weighted by their posterior model probabilities. As a concrete example, consider a Bayesian model-averaged meta-analysis with four component models formed by crossing the presence or absence of an overall effect μ with the presence or absence of between-study heterogeneity τ (Gronau et al., 2021; Bartoš et al., 2021):

- $\mathcal{H}_0^{\mathrm{FE}}$: fixed-effect null ($\mu = 0$, $\tau = 0$),
- $\mathcal{H}_0^{\mathrm{RE}}$: random-effects null ($\mu = 0$, $\tau \sim p(\tau)$),
- $\mathcal{H}_1^{\mathrm{FE}}(\gamma)$: fixed-effect alternative ($\mu \sim g(\mu \,|\, \gamma)$, $\tau = 0$),
- $\mathcal{H}_1^{\mathrm{RE}}(\gamma)$: random-effects alternative ($\mu \sim g(\mu \,|\, \gamma)$, $\tau \sim p(\tau)$),

where $g(\mu \,|\, \gamma)$ is a prior distribution on μ indexed by a hyper-parameter γ (e.g., the sd of a normal distribution) and $p(\tau)$ is a prior distribution on the heterogeneity parameter. The quantity of interest is the *inclusion Bayes factor* for μ, the evidence that μ is present, averaged over uncertainty about the heterogeneity structure. It is defined as the weighted marginal likelihood of all models that include μ divided by the weighted marginal likelihood of all models that exclude it:

$$\mathrm{BF}_{\text{incl}}(\gamma) = \frac{Z_{\text{FE}}^{(1)}(\gamma)\, p(\mathcal{H}_1^{\text{FE}}) + Z_{\text{RE}}^{(1)}(\gamma)\, p(\mathcal{H}_1^{\text{RE}})}{Z_{\text{FE}}^{(0)}\, p(\mathcal{H}_0^{\text{FE}}) + Z_{\text{RE}}^{(0)}\, p(\mathcal{H}_0^{\text{RE}})}, \tag{11}$$

where $Z_{\text{FE}}^{(1)}(\gamma)$ and $Z_{\text{RE}}^{(1)}(\gamma)$ are the marginal likelihoods of the fixed-effect and random-effects models that include μ, and $Z_{\text{FE}}^{(0)}$ and $Z_{\text{RE}}^{(0)}$ are those of the corresponding null models. The two $\mathcal{H}_1$ models share the same prior distribution on μ and hence the same sensitivity parameter γ; the denominator does not depend on γ. Because identity (7) applies separately within each model that includes μ, the marginal likelihood of each such component can be recovered across the sensitivity range from a single extended model fit and the results combined via (11).

Two implementation strategies are available. The first fits each extended model separately, one for the fixed-effect and one for the random-effects component that includes μ, and combines the resulting per-model marginal likelihood curves into $\mathrm{BF}_{\text{incl}}(\gamma)$ via (11). Each extended model fit yields a marginal likelihood *ratio* relative to its own anchor, $Z_{\text{FE}}^{(1)}(\gamma) = Z_{\text{FE}}^{(1)}(\gamma_0) \times \mathrm{MLR}_{\text{FE}}(\gamma, \gamma_0)$ and analogously for the random-effects component. To combine these in the numerator of (11), the anchor marginal likelihoods $Z_{\text{FE}}^{(1)}(\gamma_0)$ and $Z_{\text{RE}}^{(1)}(\gamma_0)$ must be available on a common absolute scale; in practice, this means computing the anchor Bayes factor of each $\mathcal{H}_1$ component against a shared reference (e.g., $\mathcal{H}_0^{\text{FE}}$) via bridge sampling or another method. Intuitively, the extended model fit for each component recovers how its marginal likelihood *changes* with γ (a ratio), but combining the fixed-effect and random-effects components in the numerator of (11) requires knowing their *absolute* marginal likelihoods at the anchor, not just their ratios. Bridge sampling against a shared null model provides this common calibration. This approach is straightforward, as each extended model is a standard single-model fit with one additional parameter, but it does not scale to settings with many components.

The second approach uses a product space sampler (Lodewyckx et al., 2011) that embeds both $\mathcal{H}_1$ components in a single MCMC run with a discrete model indicator and a shared sensitivity parameter γ. The product space approach handles the cross-calibration automatically and requires only a single extended model fit, but it demands pseudo-prior distributions for parameters absent under some components (e.g., τ under the fixed-effect model) and may mix poorly when the component models differ substantially.

2.3 Density Estimation for the Posterior of γ

The accuracy of the proposed method hinges on the quality of the posterior density estimate $\hat{p}(\gamma \mid y, \mathcal{H}_\gamma)$. In practice, the posterior distribution of γ is typically unimodal, smooth, and asymmetric, making it a comparatively easy target for density estimation. We focus on two approaches for routine use: kernel density estimation (KDE) and the importance-weighted marginal density estimator (IWMDE). Several additional methods, including logspline density estimation (Kooperberg & Stone, 1991), log-concave maximum likelihood estimation (Dümbgen & Rufibach, 2009), the asymmetric power distribution (Komunjer, 2007), and the conditional marginal density estimator (CMDE; Gelfand & Smith, 1990; Chib, 1995; Morey et al., 2011), are shown in Appendix 4 (also see Oberauer et al., 2025); however, KDE and IWMDE offer the best balance of accuracy, stability, and ease of implementation.

Kernel density estimation is the simplest nonparametric approach. A Gaussian kernel with bandwidth selected via the plug-in selector of Sheather & Jones (1991) is applied to the MCMC draws of γ, optionally boundary-corrected when γ is defined on a bounded support. In a comprehensive comparison of density estimation packages in `R`, Deng & Wickham (2014) found the `KernSmooth` package (Wand, 2025) to be both fast and accurate; we therefore use its binned kernel density estimator (Wand, 1994) as the default KDE implementation. KDE is straightforward to apply, available in any statistical computing environment, and serves as a useful baseline against which other estimators can be compared. Its main limitation is that KDE uses only the marginal posterior samples of γ, ignoring the joint draws of γ and the remaining model parameters $\boldsymbol{\theta}$. As a result, KDE cannot exploit the known functional relationship between γ and $\boldsymbol{\theta}$.

The IWMDE (Chen, 1994) exploits this additional information and is the recommended default for the proposed sensitivity analysis. The general idea is to estimate a marginal posterior density $p(\gamma^* \mid y, \mathcal{H}_\gamma)$ at any evaluation point γ^* by averaging ratios of joint posterior densities, reweighted by a conditional weighting function w, over the MCMC sample $\{(\gamma_i, \boldsymbol{\theta}_i)\}_{i=1}^n$ drawn from the full posterior under $\mathcal{H}_\gamma$:

$$\hat{p}(\gamma^* \mid y, \mathcal{H}_\gamma) = \frac{1}{n} \sum_{i=1}^{n} w(\gamma^* \mid \boldsymbol{\theta}_i)\, \frac{p(\gamma^*, \boldsymbol{\theta}_i \mid y, \mathcal{H}_\gamma)}{p(\gamma_i, \boldsymbol{\theta}_i \mid y, \mathcal{H}_\gamma)}, \tag{12}$$

where $\boldsymbol{\theta}$ denotes all remaining model parameters. When w is chosen to be the exact full conditional posterior $p(\gamma \mid \boldsymbol{\theta}, y, \mathcal{H}_\gamma)$, the IWMDE reduces to the CMDE of Gelfand & Smith (1990), which is optimal in the sense of minimizing asymptotic variance (Chen, 1994). A uniform weighting function $w = 1/(\gamma_U - \gamma_L)$ provides a consistent but higher-variance alternative that requires no knowledge of the full conditional distribution.

The crucial advantage of the IWMDE in the proposed framework is that the joint posterior ratio in (12) simplifies dramatically. Because the sensitivity parameter γ enters the model exclusively through the prior distribution on the model parameters $\boldsymbol{\theta}$, the data likelihood cancels in the ratio:

$$\frac{p(\gamma^*, \boldsymbol{\theta}_i \mid y, \mathcal{H}_\gamma)}{p(\gamma_i, \boldsymbol{\theta}_i \mid y, \mathcal{H}_\gamma)} = \frac{p(\boldsymbol{\theta}_i \mid \gamma^*)\, \pi(\gamma^*)}{p(\boldsymbol{\theta}_i \mid \gamma_i)\, \pi(\gamma_i)}, \tag{13}$$

where $p(\boldsymbol{\theta} \mid \gamma)$ is the prior density of the model parameters given γ. When the hyper-prior $\pi(\gamma)$ is uniform, the prior ratio $\pi(\gamma^*)/\pi(\gamma_i) = 1$ and the IWMDE reduces to a simple average of prior density ratios. For instance, in the Bayesian t-test with $\delta \sim$ Cauchy$(0, \gamma)$, each term is a ratio of two Cauchy density evaluations at the sampled effect size δ_i. This means the IWMDE can be computed from the MCMC draws of γ and $\boldsymbol{\theta}$ alone, without ever evaluating the data likelihood. For complex models where likelihood evaluations are expensive or not readily available in closed form, this property is a substantial practical advantage: the density estimator is effectively obtained for free as a byproduct of the extended model fit. Our empirical comparisons show that the IWMDE also yields more stable sensitivity curves than KDE, particularly in the tails of the posterior distribution and when the number of MCMC samples is moderate (see Section 3).

A simple diagnostic is to compute the sensitivity curve with both KDE and IWMDE and check for agreement; the density estimate should also be inspected visually (e.g., by overlaying it on a histogram of the posterior samples). When the sensitivity analysis involves two or more hyper-parameters simultaneously, the IWMDE generalizes directly by replacing the univariate prior density ratio with its multivariate counterpart, preserving the likelihood-free property. KDE extends to the bivariate case via the `bkde2D` function in the `KernSmooth` package, with per-dimension bandwidths selected by the same plug-in selector used in the univariate case. Because the curse of dimensionality limits nonparametric density estimation to low-dimensional $\boldsymbol{\gamma}$, sensitivity analyses involving three or more hyper-parameters may instead consider profiling (i.e., conducting one-dimensional sensitivity sweeps for each hyper-parameter while fixing the others).

2.4 Sources of Approximation Error

The identity in (7) is exact; approximation enters only through (a) the computation of the anchor Bayes factor $\mathrm{BF}_{10}(\gamma_0)$ and (b) the estimation of the posterior density ratio $p(\gamma_x \mid y, \mathcal{H}_\gamma)/p(\gamma_0 \mid y, \mathcal{H}_\gamma)$. Because these two quantities come from separate model fits, their errors are independent and propagate multiplicatively. The anchor error enters as a global scaling constant that shifts the entire sensitivity curve up or down on a log scale without distorting its shape. If bridge sampling (Gronau et al., 2017) is used for the anchor, its coefficient of variation quantifies this uncertainty.

The density ratio error reflects both MCMC sampling variability and density estimation error (see Section 2.3). What matters is the error in the *ratio* $\hat{p}(\gamma_x)/\hat{p}(\gamma_0)$: when both points lie in the high-density region of the posterior distribution, the ratio is well determined even with moderate sample sizes. When γ_x lies in the tail, the density estimate has higher relative error, but these are also the regions where the Bayes factor is extreme and the sensitivity curve is of least practical interest. The method is therefore most precise where it matters most. It is advisable to choose bounds for $[\gamma_L, \gamma_U]$ that are not excessively wider than the region where the posterior distribution has appreciable mass (see also Sec. 2.8 in Fowlie, 2024), and to check the sensitivity curve against two or more density estimation methods as a simple diagnostic.

3 Examples

In the following examples we apply the methodology to statistical scenarios of increasing complexity. First, the Bayesian t-test provides settings where we can assess the results against exact solution. Second, Bayesian model-averaged meta-analysis highlights settings where grid evaluation becomes computationally expensive. Across all examples, we run three MCMC chains with $10{,}000$ iterations each; however, the following subsections show that the IWMDE method requires roughly an order of magnitude fewer samples.

3.1 Bayesian t-test

The first example validates the proposed method against exact Bayes factors for the Bayesian two-sample t-test. Both the univariate and bivariate versions of the method are illustrated using data from the registered replication report on the facial feedback hypothesis (Wagenmakers et al., 2016). The facial feedback hypothesis holds that facial expressions can influence emotional experience; for instance, the act of smiling may make a person feel happier. Specifically, we use data from the Oosterwijk laboratory, which collected funniness ratings from 53 participants in the smile condition